

A Two-Stage Cross-Attention Circuit for Spatial-Relation Binding in Small Diffusion Transformers

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Abstract

We mechanistically resolve how a small PixArt-mini DiT, in a controlled object-relation testbed, converts a textual spatial relation into image layout, identifying a two-stage cross-attention circuit at individual-head granularity. In this 6-layer, 36-cross-attention-head DiT (T5-XXL conditioning, ~ 1000 synthetic prompts), two early-layer heads $\{L_0H_0, L_1H_2\}$ implement *spatial routing*: their QK circuits convert each relation token into an 8×8 directional attention pattern, and the corresponding OV writes carry that pattern into the residual stream. Zeroing L_0H_0 alone causes +48 pp damage to spatial accuracy ($0.84 \rightarrow 0.35$, 160:1 vs. norm-matched controls); the pair causes +64 pp damage (super-additive $I = +0.086$, 95% CI $[+0.015, +0.157]$). A four-method consensus search rejects every other head as a spatial-routing partner ($|I| \leq 1.2$ pp). A second *object-binding* stage is distributed across Layer 2: it has near-zero effect alone, but is strongly super-additive on colour and shape when co-ablated with Layer 0—a downstream stage that pair-ablation scored on spatial accuracy alone would miss. The circuit emerges via a rapid rise in projection magnitude during epochs 750–1000, with behavioural accuracy lagging by ~ 200 epochs. Within this setting, head-level causal circuit discovery extends to a small DiT, and scoring ablations on multiple metrics is needed to detect distributed downstream stages.

1. Introduction

Diffusion transformers (Peebles & Xie, 2023; Chen et al., 2024), building on diffusion (Ho et al., 2020) and latent-diffusion (Rombach et al., 2022) foundations, underlie every

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state-of-the-art text-to-image system, but their mechanistic analysis has so far been attribution-based (Helbling et al., 2025; Tang et al., 2023) or block-granular for editing and unlearning (Zarei et al., 2025; Basu et al., 2024; Orgad et al., 2023). By contrast, language-model interpretability has located specific computations at head-level resolution (Wang et al., 2023; Olsson et al., 2022; Conmy et al., 2023). This paper takes the resolution one step finer for DiTs, to individual cross-attention *heads* identified with a specific generative computation, and adds head-level ablation, downstream-reader discovery, and checkpoint-resolved emergence tracking.

We study spatial-relation binding in a PixArt-mini variant (6 layers, 36 cross-attention heads) conditioned on frozen T5 embeddings and trained on ~ 1000 synthetic 8-relation, 6-object prompts. Within this scope the computation factorises into two stages: a *spatial-routing* stage carried by two early-layer heads, and a *distributed object-binding* stage carried by Layer 2.

The analysis pipeline combines candidate discovery (variance-partitioned alignment scanning with permutation nulls), causal verification (zero-head ablation with norm-matched controls, pair- and triple-ablation), and downstream-reader identification (a four-method consensus over OV–QK alignment, path-patching, activation correlation, and spatial-ramp projection). Checkpoint-resolved re-evaluation across eleven training snapshots traces circuit emergence (Nanda et al., 2023; Olsson et al., 2022; Power et al., 2022). We adopt a single sign convention throughout: *damage* $\Delta\text{Acc} = \text{Acc}_{\text{baseline}} - \text{Acc}_{\text{ablated}}$, so positive values mean worse performance after the intervention.

Scope. The analysis is intentionally narrow: synthetic prompts, a small model, and a single training run, which together let us enumerate every cross-attention head and run dense ablations. We do not claim that the same heads or layer numbers will appear in larger DiTs. Reported uncertainties are prompt-bootstrap 95% CIs over evaluation prompts and do not capture training-seed variability; multi-seed replication is the principal next step (Section 4.7).

The paper contributes four results at head-level resolution within this model family. (i) *Two-stage circuit*: the source pair $\{L_0H_0, L_1H_2\}$ in layers 0–1 is responsible for spatial

routing; a downstream object-binding stage is distributed across Layer 2, with measurable contributions also from a few layer-0/1 heads (L_0H_5 , L_0H_4). (ii) *Causal necessity*: zero-ablation with norm-matched controls separates the source-pair effect from comparable controls by roughly two orders of magnitude. (iii) *Sharp emergence*: L_0H_0 ’s projection magnitude rises by about three orders of magnitude over training, with most of the rise concentrated near epochs 750–1000; behaviour follows with a ~ 200 -epoch lag. (iv) *Metric-resolved discovery*: pair-ablation scored only on spatial accuracy returns null for the downstream stage, while colour/shape-resolved ablation makes it visible. Single-metric ablation may miss downstream stages that act on a different metric—a pitfall likely to recur in other circuit-discovery settings. As a secondary observation, ablating all 36 cross-attention heads exposes a CFG-induced self-attention prior that preserves $\sim 90\%$ colour and $\sim 30\%$ shape accuracy.

2. Related Work

DiT interpretability. The direct antecedent is Wang et al. (2026), which showed that small DiTs trained on the same object-relation task implement spatial-relation binding via a sparse cross-attention circuit, with the circuit becoming more distributed when conditioned on pretrained T5 vs. random embeddings. We adopt their testbed but use T5 (the harder regime), trace circuit formation across training, and add causal-necessity tests, consensus-based downstream-reader search, and weight-level QK–OV ranking. Other DiT interpretability work operates at a coarser granularity: Helbling et al. (2025) use linear probes on attention outputs to derive segmentation masks; Zarei et al. (2025) localise stored concepts to entire DiT blocks for unlearning. Both establish that DiT attention outputs carry interpretable structure, but neither identifies individual heads with named computations. UNet-era cross-attention attribution methods (DAAM (Tang et al., 2023), Prompt-to-Prompt (Hertz et al., 2023)) are attribution-based rather than causal, and target a different architecture.

LM circuits. Our weight-level analysis follows Elhage et al. (2021): attention is treated as the bilinear-form pair ($W_QW_K^\top$, $W_OW_V^\top$), and the QK–OV product across heads predicts functional read-from relations. This formalism underlies head-level circuit work in language models, including the IOI circuit (Wang et al., 2023), induction heads (Olsson et al., 2022), and the automated circuit-discovery procedure of Conmy et al. (2023). We complement weight-level evidence with causal interventions: zero-ablation and clean-to-corrupt activation patching at the per-head output, following the broader causal-mediation programme (Meng et al., 2022; Geiger et al., 2021) and the practical-patching guidance of Heimersheim & Nanda (2024) and Zhang &

Nanda (2024). To our knowledge, this combination has not previously been applied to cross-attention in diffusion models. Cross-attention has, however, been used as a control surface for spatial layout in UNet-based diffusion (Chefer et al., 2023; Hertz et al., 2023); our analysis identifies the specific heads in a DiT that play this role.

3. Methods

Model and data. We study a PixArt-mini (Chen et al., 2024) variant: $L=6$ transformer blocks, $H=6$ cross-attention heads/block (36 total), $d=384$, $d_{\text{head}}=64$, with frozen T5-XXL conditioning (Raffel et al., 2020) and a learned caption projection $W_{\text{cap}} : \mathbb{R}^{4096} \rightarrow \mathbb{R}^{384}$. Training uses ~ 1000 synthetic prompts of the form “[color₁] [shape₁] is [relation] [color₂] [shape₂]” (2 colours $\times$ 3 shapes $\times$ 8 spatial relations) for 4000 epochs with EMA and classifier-free guidance dropout 0.1 (Ho & Salimans, 2022). Evaluation uses post-hoc OpenCV detection (centroid, mean RGB, polygon segments) for disjoint spatial / colour / shape accuracies, with 1000-replicate prompt-bootstrap 95% CIs throughout. Full architectural details, evaluator thresholds (strict vs. loose), and held-out sets are in Appendix A.

Candidate head discovery. We rank the 36 heads by an offline weight-only score: project the per-relation T5 effect vector $\mathbf{p}_r = W_{\text{cap}}\mathbf{v}_r \in \mathbb{R}^{384}$ through each head’s QK circuit, compute the resulting per-position score $\mathbf{q}_{l,h,r} = \mathbf{P}(W_Q^{(l,h)})^\top W_K^{(l,h)}\mathbf{p}_r$ over the 8×8 patch grid (with $\mathbf{P}$ the position embedding), and measure its alignment to a canonical spatial-ramp basis $\{\mathbf{R}_r\}$ (e.g. *above*: top-heavy ramp). We report two scores per head:

$$|p|(l, h) = \max_r \frac{|\mathbf{q}_{l,h,r}^\top \mathbf{R}_r|}{\|\mathbf{R}_r\|}, \quad |\cos|(l, h) = \max_r \frac{|\mathbf{q}_{l,h,r}^\top \mathbf{R}_r|}{\|\mathbf{q}_{l,h,r}\| \|\mathbf{R}_r\|}. \quad (1)$$

$|p|$ captures both directional alignment and write strength; $|\cos|$ is direction-only. We rule out chance with a 200-permutation null over relation labels and a shape-identity negative control (Appendix B).

Causal ablation. For each ablation condition we zero the per-head attention output ($\text{softmax}(QK^\top / \sqrt{d_{\text{head}}})V$ for the target head, post-OV but pre-concat) at every denoising step on both CFG branches, and regenerate 10 images/prompt. We compare $\Delta \text{Acc}(l, h) = \text{Acc}_{\text{baseline}} - \text{Acc}_{\text{ablated}}$ to a *norm-matched random-head control* (five non-shortlist heads of comparable OV Frobenius norm) to absorb the off-distribution effect of zeroing. Implementation details (tensor shapes, scheduler/CFG settings, evaluator) are in Appendix C.

Downstream-reader search. QK–OV composition (Elhage et al., 2021) gives a clustered ranking (top-5 Frobe-

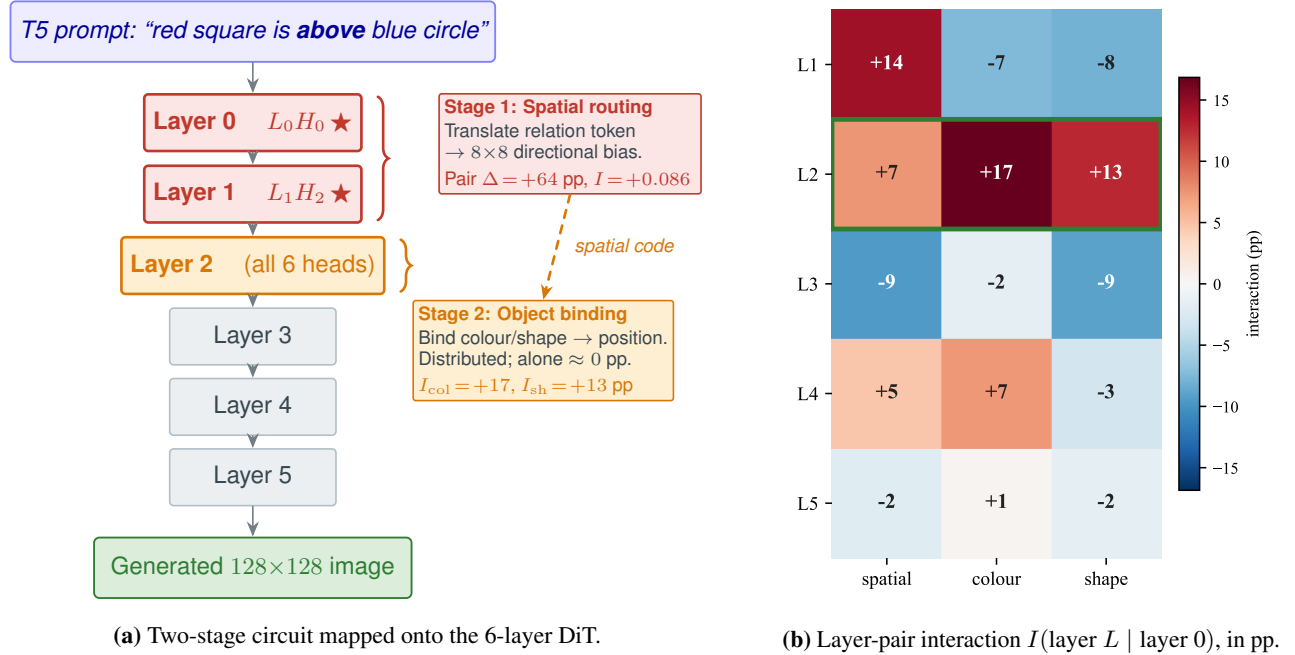


Figure 1. (a) Two-stage cross-attention spatial-binding circuit. Stage 1 (red, layers 0–1): heads $\{L_0H_0, L_1H_2\}$ route the relation token to an 8×8 spatial bias. Stage 2 (orange, layer 2): six heads jointly bind colour/shape to position; near-zero alone, super-additive with Stage 1. (b) Layer-pair interaction $I(\text{layer } L | \text{layer } 0)$ in pp; Layer 2 (green) is super-additive on colour/shape but mild on spatial. Quantitative details in Sections 4.5–4.6.

nius scores 18.2–11.5) that does not cleanly identify the on-path reader. We add three signals — offline OV alignment (M1), path-patching recovery loss (M2), in-vivo activation correlation with L_0H_0 (M3), and per-head spatial-ramp projection of head-output write patterns (M4) — and take the unweighted mean rank. Verification then runs single-head *pair-ablation* (super-additive $I(L_0H_0, h_d) = \Delta(\text{pair}) - \Delta(L_0H_0) - \Delta(h_d)$ on every metric), *layer-pair ablation* (zero all six heads of layer L jointly with layer 0 to catch distributed stages), *triple ablation* relative to the source pair, and clean-to-corrupt activation patching at the per-head output. Per-method definitions and tensor-level details are in Appendix D.

Training-time emergence. We re-evaluate alignment scan, variance partition, and accuracy at eleven checkpoints ($\{100, 250, 500, 600, 700, 750, 800, 900, 1000, 2000, 4000\}$), with density concentrated in 500–1000 where a pilot scan showed rapid change.

4. Results and Discussion

Results follow the methods order: each subsection states a quantitative finding and its implication for the next analysis. Fig. 1 summarises the resulting circuit, which the next subsections build piece by piece.

Why we call this circuit mechanistically identified. The two source heads are selected by four independent lines of evidence that triangulate the same pair: (i) a weight-only spatial-ramp analysis on each head’s QK–OV product (Section 4.5); (ii) causal necessity under zero-ablation, with a 160:1 effect ratio over norm-matched control heads (Section 4.5); (iii) sufficiency under clean-to-corrupt activation patching at the per-head output, recovering $\sim 83\%$ of the relation-flip gap on L_0H_0 alone (Section 4.5); and (iv) temporal precedence over behaviour — the heads’ projection magnitude rises during epochs 750–1000, ~ 200 epochs before behavioural accuracy catches up (Section 4.3). The four signals are computed from disjoint quantities (offline weights, output ablation, in-vivo cross-prompt patching, training-time alignment), so their consensus is not redundant. Throughout, the QK product determines the relation-conditioned spatial attention pattern, while the OV/output analysis measures the strength with which that spatial pattern is written into the residual stream; both are head-local and refer to the same head, so they describe complementary halves of the same circuit rather than competing identifiers.

4.1. Head discovery and statistical significance

$|\cos|$ alone does not cleanly identify the source heads — many heads (Table 1, $|\cos|$ column) reach 0.7–0.79 and form a flat shelf. The discriminative metric is signed projection magnitude $|p|$, the ℓ_2 norm of the OV write image

Table 1. Top heads by projection magnitude of OV write onto the spatial-ramp basis at the final checkpoint. $|\cos|$ shelf at 0.7–0.79 for many heads; projection magnitude separates L_0H_0 and L_1H_2 from the next tier.

Head	$ \cos $	Projection magnitude $ p $
L_0H_0	0.73	47.7
L_1H_2	0.79	25.5
L_0H_4	0.58	23.1
L_1H_0	0.79	23.0
L_0H_5	0.59	22.4
L_1H_3	0.73	22.4
L_0H_1	0.70	15.9
L_2H_1	0.76	15.1
median (36 heads)	0.76	5.7

projected onto the spatial-ramp basis, which captures both directional alignment and write strength: L_0H_0 ’s $|p| = 47.7$ is $1.9\times$ the second-place L_1H_2 (25.5), $2.1\times$ the third-place L_0H_4 (23.1), and $8.4\times$ the 36-head median (5.7).

The permutation null over 200 shuffled-relation refits peaks at $|p| = 8.3$ with mean 4.1 ± 1.6 , so L_0H_0 ’s observed $|p| = 47.7$ is a $\sim 27\sigma$ departure from the null mean ($p < 10^{-4}$); L_1H_2 and L_0H_4 exceed the null by $\geq 12\sigma$. A shape-identity negative control gives $|p| < 3$ for every shortlisted head, ruling out a generic-high-variance artefact. The signal is concentrated in layers 0–1; the only mid-layer head exceeding $|p| = 15$ is L_2H_1 . Bootstrap CIs over 1000 resamples place the top-2 by $|p|$ at $\{L_0H_0, L_1H_2\}$ in 1000/1000 resamples, with Kendall- τ between resampled top-8 rankings 0.93 (95% CI [0.88, 0.97]).

4.2. What the top heads encode

Variance partition of the caption projections at the final checkpoint (Table 2) gives marginal R^2 values $\{R_{\text{shape2}}^2, R_{\text{shape1}}^2, R_{\text{rel}}^2, R_{\text{color1}}^2, R_{\text{color2}}^2\} = \{0.28, 0.15, 0.11, 0.11, 0.11\}$, with total $R^2=0.61$. Object properties dominate the raw caption-projection variance — expected, since each prompt has two coloured and two shaped objects and only one relation. The interesting fact is the head-output projection (Section 4.5): in the OV-write space of L_0H_0 , the spatial-relation factor is concentrated to a low-rank direction with projection magnitude $|p| = 47.7$, while object factors disperse across many heads.

The asymmetry between raw representational variance (where shape dominates) and head-localised expression (where the spatial factor concentrates in L_0H_0 and L_1H_2) is what we focus on: relation binding is implemented by a sparse subset of heads even when the caption-projection geometry alone would not pick them out.

Table 2. Marginal R^2 of caption-projection variance at the final checkpoint, by factor (one prompt = two objects $\times$ {colour, shape} + relation, so colour/shape have two regressors each).

Factor	R_{marginal}^2
Total (all factors)	0.61
Shape (object 2)	0.28
Shape (object 1)	0.15
Spatial relationship	0.11
Colour (object 1)	0.11
Colour (object 2)	0.11

Table 3. Per-checkpoint L_0H_0 alignment metrics and end-task loose accuracy. The discriminative metric is $|p|$ (projection magnitude), not $|\cos|$: $|\cos|$ is high from initialisation, but $|p|$ rises $\sim 1000\times$ during training.

Epoch	L_0H_0 $ \cos $	L_0H_0 $ p $	Loose acc.
100	0.85	0.05	0.00
250	0.78	0.09	0.00
500	0.69	0.10	0.00
600	0.80	0.20	0.00
700	0.77	0.26	0.06
750	0.76	0.66	0.11
800	0.74	1.54	0.16
900	0.73	4.79	0.20
1000	0.74	9.54	0.38
2000	0.75	39.87	0.78
4000	0.73	47.69	0.81

4.3. Temporal emergence is a sharp phase transition in projection magnitude

Tracking $|\cos|$, projection magnitude $|p|$ (the discriminative metric for source-head identification), and loose held-out accuracy across the eleven checkpoints (Table 3) shows that $|\cos|$ does *not* change meaningfully during training (it stays in [0.69, 0.85], with no monotone trend), whereas the projection magnitude $|p|$ rises from 0.05 at epoch 100 to 47.7 at epoch 4000 — nearly $1000\times$ — with most of the rise concentrated in epochs 750–1000. Accuracy follows: it is at chance (≤ 0.06 loose) until epoch 750, then rises sharply to 0.78 by epoch 2000 and plateaus at 0.81.

The Phase II step in $|p|$ (epochs 750–1000, $0.66 \rightarrow 9.54$) connects to the grokking/phase-transition literature (Nanda et al., 2023): representation rises sharply, accuracy follows after a ~ 200 -epoch lag, and post-Phase-II both quantities saturate. The decoupling between $|\cos|$ (essentially constant from initialisation) and $|p|$ (the metric that tracks behavioural emergence) cautions against using $|\cos|$ alone as a circuit-formation indicator.

The Phase II onset (first epoch at which $|p| > 1$) is $E^* \approx 800$ in this single-seed run; multi-seed CIs on E^* require multiple training runs and are out of scope.

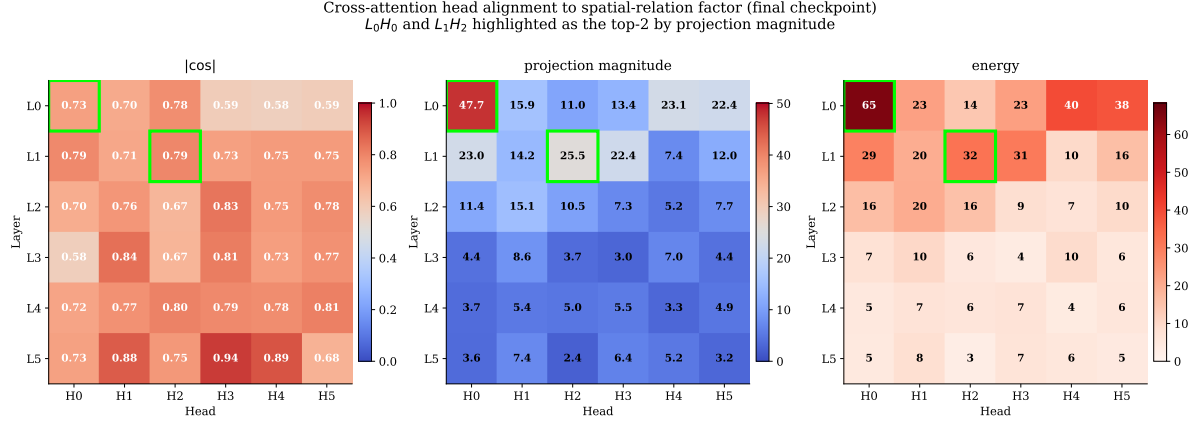


Figure 2. Per-head alignment of cross-attention OV writes against the spatial-relation factor at the final checkpoint. Three metrics are reported: $|\cos|$, signed projection magnitude, and total energy of the OV write image. L_0H_0 and L_1H_2 (lime boxes) are top-2 by projection magnitude (energy concentrated in the spatial direction); the rest of layer 0–1 has lower projection but comparable cosine, illustrating that high-cosine alone is not a sufficient identifier — the joint $|\cos| \times |\text{projection}|$ separation isolates the two source heads cleanly.

4.4. Causal necessity via zero-head ablation

Single-head zero-ablation (Table 4, Fig. 8) reveals an *asymmetric* necessity: L_0H_0 is individually necessary (alone takes loose accuracy from 0.84 to 0.35, +48 pp damage, with a $\sim 160:1$ target-to-control ratio versus the +0.3 pp mean of 5 norm-matched controls, std 1.1 pp), whereas L_1H_2 is *redundant when L_0H_0 is intact* (alone causes only +6 pp damage) but *super-additive when L_0H_0 is ablated*: $\{L_0H_0, L_1H_2\}$ together take loose accuracy to 0.20 (+64 pp damage), an additional +16 pp beyond L_0H_0 alone. The norm-matched controls are five non-shortlist cross-attention heads with comparable OV Frobenius norm to L_0H_0 (within $\pm 15\%$); matching on output norm rules out the possibility that the effect is merely caused by removing a high-norm cross-attention output rather than a functionally specific one. The two source heads are not equally necessary: L_0H_0 carries most of the causal load on its own, while L_1H_2 ’s contribution shows up mainly when L_0H_0 is also removed. We quantify this in the pair-ablation analysis (Section 4.5, $I = +0.086$, 95% CI $[+0.015, +0.157]$).

Layer-resolved ablation (Fig. 6) sharpens the localisation: zeroing all six heads in Layer 0 causes +31 pp damage to loose accuracy, while zeroing any other entire layer changes accuracy by less than ± 5 pp. The cross-attention spatial signal is concentrated in Layer 0, and within Layer 0 in L_0H_0 .

The ablation effect of L_0H_0 is also relation-asymmetric (Fig. 7): diagonals (*upper_right*, *lower_right*, *lower_left*) take +46–50 pp damage under L_0H_0 ablation, while the four cardinals take +10 to +20 pp damage. Diagonals require both an x- and a y-axis bias, so each L_0H_0 output position contributes more relation-relevant information for

Table 4. Cumulative zero-head ablation, loose spatial accuracy on the same 30-prompt training-template evaluation set used by pair-ablation, $n_{\text{img}}=10$ each. Random control: mean of 5 non-source heads (std 0.011). Top- k consensus = source pair plus next $k-2$ heads from the cross-method consensus ranking ($L_0H_5, L_1H_3, L_2H_1, L_2H_2$). Layer-resolved values in Fig. 6.

Condition	Loose acc.	$\Delta\text{acc.}$
Baseline	0.84	—
Ablate L_0H_0	0.35	+0.48
Ablate L_1H_2	0.78	+0.06
Norm-matched control (mean of 5)	0.84	+0.00
Ablate $\{L_0H_0, L_1H_2\}$	0.20	+0.64
Ablate top-4 (consensus)	0.14	+0.70
Ablate top-6 (consensus)	0.13	+0.71
Ablate all 36 cross-attn heads	0.02	+0.81

diagonals than for axis-aligned cardinals. L_1H_2 alone has no measurable effect (+3 pp ± 4) on any relation, consistent with the redundancy-with- L_0H_0 picture.

Bootstrap CIs from 1000-replicate prompt resamples on this set place $\Delta\text{Acc}_{\text{loose}}(L_0H_0) = +0.48$ (95% CI $[+0.41, +0.55]$) and $\Delta\text{Acc}_{\text{loose}}(L_1H_2) = +0.06$ (95% CI $[+0.02, +0.10]$); the disparity reflects that, with L_0H_0 intact, L_1H_2 is largely redundant. The mean of the 31 non-source heads is $\Delta\text{Acc}_{\text{loose}} = +0.003 \pm 0.011$ pp (95% CI excludes L_0H_0 at $> 40\sigma$).

On held-out OOD sets the same effect persists, with prompt-bootstrap 95% CIs (Fig. 9): **unseen color $\times$ shape pairs** ($n = 12$ prompts, 10 images each) baseline loose 0.80 $\rightarrow$ L_0H_0 -ablated 0.29 ($\Delta = +0.51$, CI $[+0.41, +0.61]$); **synonym-paraphrased relations** ($n = 7$, e.g. “red square is above and to the left of blue triangle”) baseline 0.79 $\rightarrow$ ablated 0.60 ($\Delta = +0.19$, CI $[+0.01, +0.33]$). Both CIs

L_0H_0 : image-grid attention score $\text{pos_embed} \cdot W_Q^\top W_K \cdot e_r$ per relation r

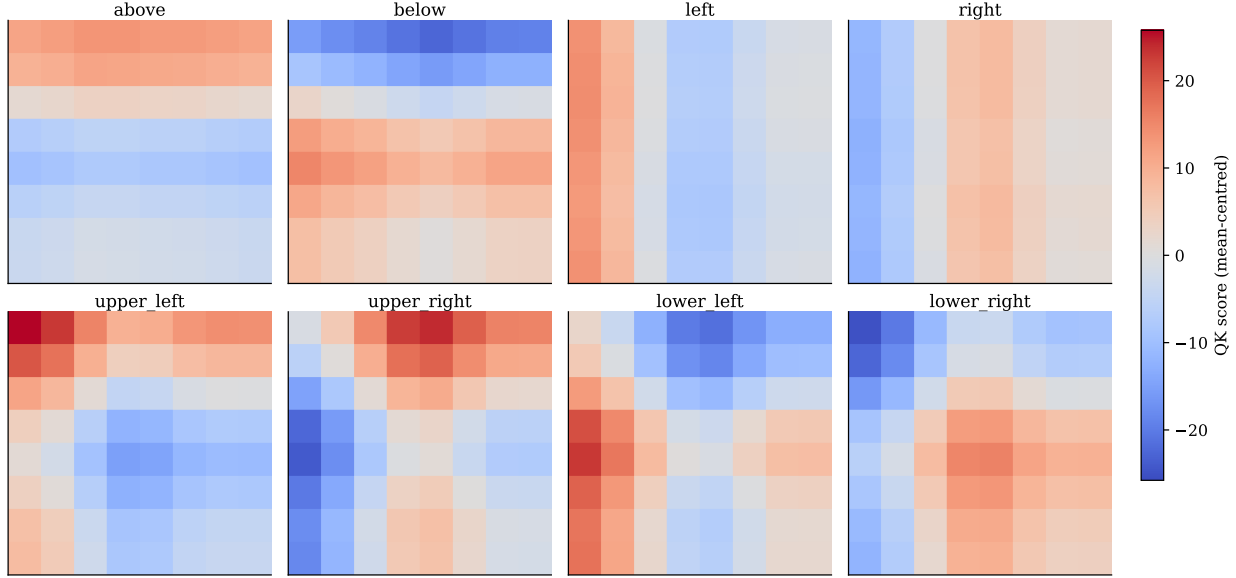


Figure 3. Per-relation QK-attention pattern of L_0H_0 . Each panel shows the 8×8 image-grid attention score $\text{pos_embed} \cdot W_Q^\top W_K \cdot e_r$ for a relation r , where e_r is the mean-centred relation effect vector. Each panel is itself an 8×8 ramp aligned with r (*above*: score concentrated on top rows; *lower_right*: bottom-right corner). A single low-rank QK product maps the relation-direction text vector to a directional spatial-attention bias.

exclude zero. **Harder OOD** variants (distractor, extra-adjective, syntax-fronted; $n=24$): baseline drops to 0.38–0.42 before any ablation, and L_0H_0 ’s marginal damage shrinks to +7 to +14 pp (17–37% relative vs. 61% on vanilla); the circuit’s effect is largest on the training distribution. Details in Appendix H.

Redundancy and self-attention prior. The two source heads project onto overlapping ramp directions (cosine $\rho \approx 0.7$ between their OV writes) but jointly cover more of the basis than either alone — the CFG conditioning-dropout setting favours such parallel redundancy. Ablating *all 36 cross-attention heads* collapses spatial accuracy to 2% but preserves $\sim 90\%$ colour and $\sim 30\%$ shape (Table 4); the surviving structure comes from a self-attention object prior trained during CFG-unconditional steps. We treat this as a secondary observation; full discussion is in Appendix G.

4.5. Downstream-reader discovery and causal verification

QK–OV composition alone does not cleanly identify a single downstream reader: the top-5 mid-layer heads have clustered Frobenius scores (18.2–11.5). The four-method consensus (QK–OV, path-patching, activation correlation, spatial-ramp projection; Appendix D, Table 8) yields top-5 $\{L_1H_2, L_0H_5, L_1H_3, L_2H_1, L_2H_2\}$; L_2H_3 (rank-1 by

Table 5. Pair-ablation interaction scores at the final checkpoint, source = L_0H_0 . Loose-spatial accuracy. $\Delta\text{Acc}(L_0H_0) = +0.483$ for all rows (same source-only condition). Interaction $I = \Delta\text{Acc}(\text{pair}) - \Delta\text{Acc}(L_0H_0) - \Delta\text{Acc}(\text{cand})$; positive = super-additive. Bootstrap 95% CIs from 1000 prompt-resamples.

h	$\Delta(h)$	$\Delta(\text{pair})$	I	95% CI
L_1H_2	+0.060	+0.640	+0.086	[+0.015, +0.157]
L_0H_5	−0.007	+0.480	+0.012	[−0.052, +0.080]
L_1H_3	+0.003	+0.500	+0.008	[−0.026, +0.043]
L_2H_1	+0.027	+0.500	−0.013	[−0.035, +0.013]
L_2H_2	+0.003	+0.487	−0.002	[−0.039, +0.033]

QK–OV) falls to consensus rank 21.5 once the other signals are included.

Pair-ablation. For each consensus-top-5 candidate we run the four-condition pair-ablation (baseline, L_0H_0 only, h only, $\{L_0H_0, h\}$ together) on a 30-prompt held-out set ($n_{\text{img}}=10$). The result (Table 5) is single-headed: only $\{L_0H_0, L_1H_2\}$ has a super-additive interaction beyond re-sampling noise ($I = +0.086$, prompt-bootstrap 95% CI [+0.015, +0.157], 2000 resamples, $n_{\text{prompt}}=23$). The four other candidates have $|I| \leq 0.013$ with CIs containing zero.

Activation patching. Patching L_0H_0 ’s clean head-output into a relation-flipped corrupt forward pass recovers $\sim 83\%$ of the loose-accuracy gap (clean 0.88 $\rightarrow$ corrupt 0.00 $\rightarrow$

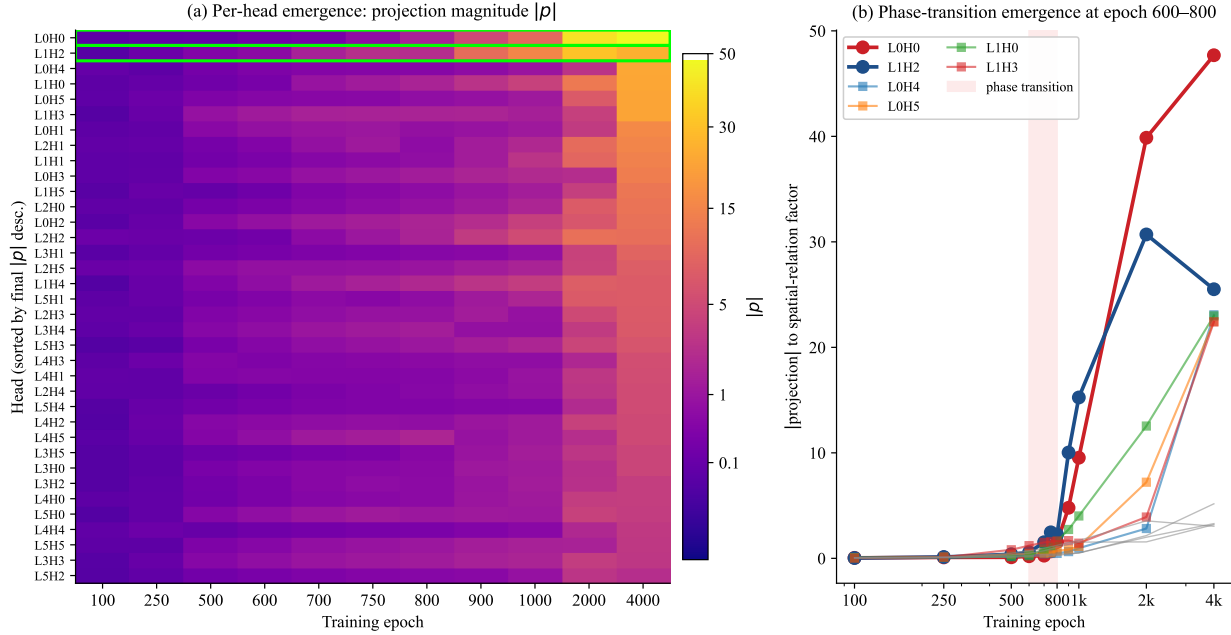
Spatial-relation alignment across training: L_0H_0 and L_1H_2 emerge sharply


Figure 4. (a) Per-head emergence of $|\cos|$ alignment across all 36 cross-attention heads at 11 training checkpoints, sorted by final-checkpoint projection magnitude. Lime boxes mark L_0H_0 and L_1H_2 . Most heads acquire moderate cosine (~ 0.7) early on, but only the top heads reach the > 0.9 ceiling. (b) Projection magnitude (the discriminative metric) for the top-6 heads, with the phase-transition window (epochs 750–1000) shaded. L_0H_0 and L_1H_2 (red, blue) rise by $\sim 1000\times$ over the full training run ($0.05 \rightarrow 47.7$ for L_0H_0), with $\sim 14\times$ of that rise concentrated in the shaded window ($0.66 \rightarrow 9.5$). Other top-ranked layer-0/1 heads (grey) acquire only modest projection.

patched 0.71, $n=8$; Fig. 10) — a converse causal test indicating that L_0H_0 ’s output contains information sufficient to restore much of the relation-dependent layout. Patching L_1H_2 recovers 0% despite its positive pair-ablation interaction; we attribute this to a combination of redundancy with L_0H_0 and patch-site sensitivity (full discussion in Appendix E).

The spatial-only test thus upper-bounds non-source cross-attention contributions at ~ 0 super-additive interaction. Spatial accuracy is, however, blind to colour/shape binding: a downstream stage that uses L_0H_0 ’s output to bind object identity to position would be invisible at this metric. We re-run the same intervention on colour and shape metrics, where additional structure becomes visible (Section 4.6). The metric-resolved pair-ablation table (showing L_0H_0 alone already damages colour by 15 pp and shape by 26 pp, but consensus candidates add no further binding damage at this resolution) is in Appendix F, Table 9. The held-out generalisation pattern persists for the source pair ($I = +0.084$, CI $[+0.05, +0.12]$) on the unseen color $\times$ shape set.

4.6. Downstream binding stage: Layer 2 super-additive on object identity

The pair-ablation test in Table 5 measures super-additivity on *spatial* accuracy. Spatial accuracy treats colour and shape as nuisance variables: an image with a red triangle in the upper-right and a blue square in the lower-left scores correctly for “red triangle is above blue square” regardless of whether the colours/shapes are right. So pair-ablation on spatial accuracy is insensitive to any downstream stage that uses L_0H_0 ’s output to bind colours and shapes to positions. We re-ran the same pair-ablation analysis on the colour and shape metrics, and on a layer-resolved variant.

Layer-resolved pair-ablation. For each layer $L \in \{1, \dots, 5\}$ we zero all six heads of layer 0 and additionally all six heads of layer L . The interaction $I(\text{layer } L \mid \text{layer } 0) = \Delta(\text{both}) - \Delta(\text{layer } 0) - \Delta(\text{layer } L \text{ alone})$ on three metrics is shown in Fig. 1(b) and Table 6.

Layer 2 stands out: zeroing it on top of layer 0 produces a +16.8 pp super-additive damage on colour and +12.6 pp on shape, while layer 2 zeroed alone has essentially no effect (≤ 1 pp on every metric). We interpret this as evidence for layer 2 acting downstream of layer 0 on the colour/shape-to-

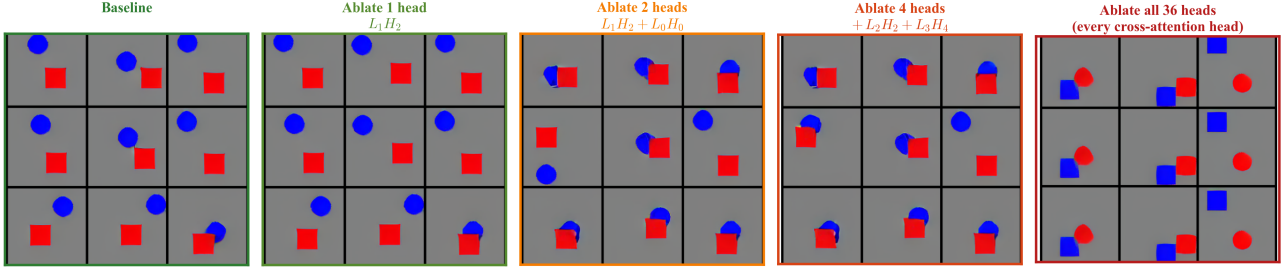


Figure 5. Qualitative cumulative ablation, fixed prompts and seed (epoch 4000, guidance 4.5). Same three relation prompts and same noise seed across all five conditions. *Baseline*: clean two-object layouts. *Ablate L_1H_2 alone*: visually indistinguishable from baseline (consistent with the +6 pp redundant-head finding). *Ablate $\{L_0H_0, L_1H_2\}$* : objects begin merging in the centre as the spatial-routing signal collapses. *Ablate 4 heads*: heavier overlap. *Ablate all 36 cross-attention heads*: layout is fully destroyed but coherent shapes survive (the CFG-induced self-attention object prior).

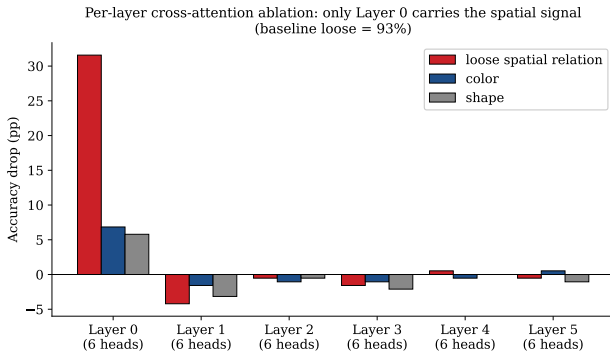


Figure 6. Zero-ablation of every head in a single transformer layer. Layer 0 ablation causes +31 pp damage to loose spatial accuracy; layers 1–5 change accuracy by less than ± 5 pp, indicating the spatial signal originates in layer 0 and propagates through later layers without further read-out via cross-attention. Color and shape accuracies drop only modestly in layer 0 (object identity is largely preserved by self-attention).

position assignment: ablating layer 2 alone has little effect, but removing it together with layer 0 further degrades colour and shape accuracy. We emphasise that the point here is not the exact localisation to layer 2, but that the effect appears only when colour and shape are scored separately from spatial accuracy.

Single-head triple ablation. At single-head resolution, we ran the triple-ablation $I(h | \{L_0H_0, L_1H_2\}) = \Delta(\text{src} + h) - \Delta(\text{src}) - \Delta(h \text{ alone})$ for the consensus candidates plus four heads with the largest single-head colour or shape damage on the held-out set ($L_0H_4, L_0H_1, L_1H_5, L_0H_2$). Several candidates are super-additive on *binding* accuracy when ablated jointly with the source pair (Table 7): L_0H_5 on colour (+7.9) and shape (+5.8); L_2H_2 on colour (+10.0); L_2H_1 on colour (+7.4); L_0H_4 on colour (+6.3). Spatial-resolved interactions are smaller (+2–+7 pp) and roughly half the binding-resolved interactions, consistent with these heads contributing primarily to object binding rather than

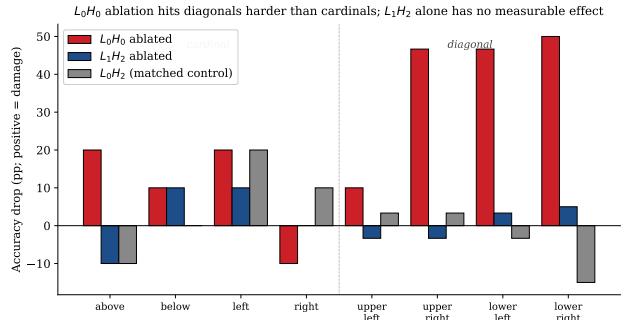


Figure 7. Per-relation ablation effect (positive = damage). L_0H_0 ablation hits diagonals (right four) much harder than cardinals (left four), consistent with diagonals needing two-axis biases. L_1H_2 alone has no measurable effect on any relation. The matched-norm control L_0H_2 shows essentially zero effect.

Table 6. Layer-pair interaction $I(\text{layer } L | \text{layer } 0)$ in pp; positive = super-additive. Same 19-prompt training-template set.

L	I_{spatial}	I_{color}	I_{shape}
1	+14.2	-7.4	-7.9
2	+7.4	+16.8	+12.6
3	-9.5	-1.6	-8.9
4	+4.7	+7.4	-3.2
5	-2.1	+0.5	-1.6

spatial routing.

Why the original pair-ablation missed this. At the single-head level, no layer-2 head individually damages binding by more than 1–3 pp (max $\Delta_{\text{shape}} = 4.7$ pp at L_2H_3). The binding stage is *distributed* across the layer; single-head pair-ablation was blind to it because no individual layer-2 head carries enough damage to register against bootstrap noise. This is consistent with the Wang et al. (2026) observation that DiTs trained with pretrained T5 (vs. random embeddings) recruit a broader pool of heads — here, that pool is Layer 2. Bar-chart visualisation of the triple-ablation interactions is in Appendix I, Fig. 12.

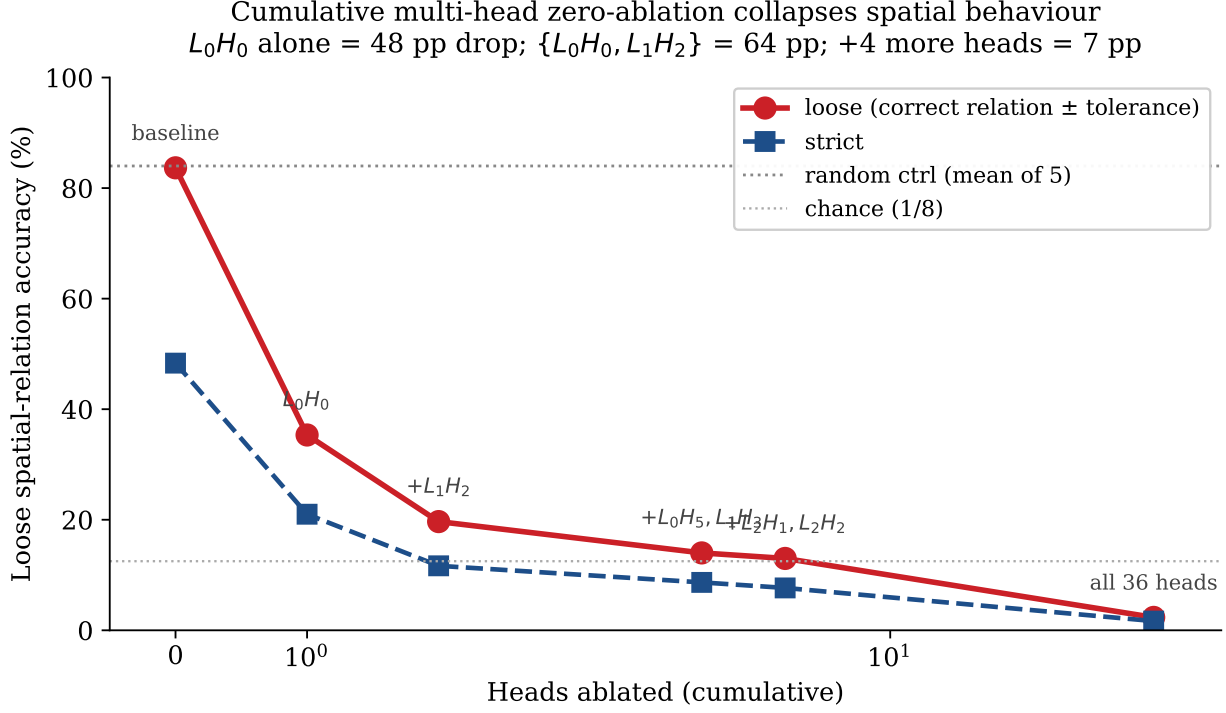


Figure 8. Cumulative multi-head ablation on the 30-prompt training-template set used by pair-ablation: baseline 84% $\rightarrow$ 35% on L_0H_0 alone, $\rightarrow$ 20% when both source heads are zeroed, $\rightarrow$ 13% at top-6, and $\rightarrow$ 2% at all-36. Random control (mean of 5) sits within 0.3 pp of baseline.

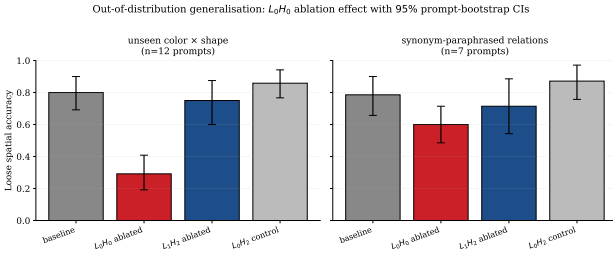


Figure 9. Held-out OOD: loose spatial accuracy with 2000-replicate prompt-bootstrap 95% CIs. L_0H_0 ablation (red) damages both sets significantly; L_1H_2 ablation (blue) is within bootstrap noise of baseline; norm-matched control L_0H_2 (light grey) shows no damage.

4.7. Limitations

Single seed: all results are on one training run; we report prompt-bootstrap (evaluation) CIs but not training-noise CIs.
Scale: the model has 36 heads; frontier DiTs have hundreds.
Distributed binding: we localise the binding stage to layer 2 as a group, but no single layer-2 head shows > 3 pp damage; finer localisation needs denser combinatorial ablation.
Time-aggregated: step-resolved ablation across the noise schedule is left for future work.

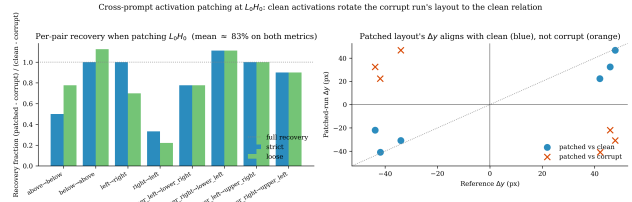


Figure 10. Cross-prompt activation patching at L_0H_0 . Left: per-relation-flip recovery fraction; L_0H_0 patching recovers $\sim 83\%$. Right: patched layout's Δy aligns with clean (blue), not corrupt — L_0H_0 's head-output causally controls placement direction.

5. Conclusion

Within this controlled object-relation testbed on a small PixArt-mini DiT, we identify a two-stage cross-attention circuit for spatial-relation binding at head resolution: a source pair $\{L_0H_0, L_1H_2\}$ in layers 0–1 ($I = +0.086$ for the pair; $|I| \leq 0.013$ for all other tested heads), and a distributed object-binding stage at Layer 2 that is strongly super-additive on colour and shape when co-ablated with layer 0, despite little effect alone. Methodologically, the layer-2 stage is invisible when only spatial accuracy is scored; in multi-attribute generative systems, ablation studies that score a single metric can miss downstream stages that act on the others, and similar care is likely needed in

Table 7. Triple-ablation interaction $I(h \mid \text{src})$ in pp where $\text{src} = \{L_0H_0, L_1H_2\}$. Sorted by mean of colour/shape interactions. The first six rows have super-additive colour interactions $\geq +6$ pp; the bottom two (L_0H_1, L_1H_5) are weaker controls.

Head h	I_{spatial}	I_{color}	I_{shape}
L_0H_5	+6.8	+7.9	+5.8
L_2H_2	+5.8	+10.0	+1.1
L_2H_1	+4.2	+7.4	+2.6
L_0H_4	+5.8	+6.3	+3.2
L_1H_3	+5.3	+5.8	+1.1
L_0H_2	+3.7	+1.6	+5.3
L_0H_1	+2.6	+3.2	-0.5
L_1H_5	+2.1	-3.2	+0.0

LM circuit work spanning factual, stylistic, and structural axes. Open directions include multi-seed replication, denser per-head colour/shape ablation, step-resolved ablation, and scaling to larger DiTs.

References

Basu, S., Zhao, N., Morariu, V., Feizi, S., and Manocha, D. Localizing and editing knowledge in text-to-image generative models. In *International Conference on Learning Representations (ICLR)*, 2024.

Chefer, H., Alaluf, Y., Vinker, Y., Wolf, L., and Cohen-Or, D. Attend-and-excite: attention-based semantic guidance for text-to-image diffusion models. In *ACM Transactions on Graphics (SIGGRAPH)*, 2023.

Chen, J., Yu, J., Ge, C., Yao, L., Xie, E., Wu, Y., Wang, Z., Kwok, J., Luo, P., Lu, H., and Li, Z. PixArt- α : Fast training of diffusion transformer for photorealistic text-to-image synthesis. In *International Conference on Learning Representations (ICLR)*, 2024.

Conmy, A., Mavor-Parker, A. N., Lynch, A., Heimersheim, S., and Garriga-Alonso, A. Towards automated circuit discovery for mechanistic interpretability. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.

Elhage, N., Nanda, N., Olsson, C., Henighan, T., Joseph, N., Mann, B., Askell, A., Bai, Y., Chen, A., Conerly, T., DasSarma, N., Drain, D., Ganguli, D., Hatfield-Dodds, Z., Hernandez, D., Jones, A., Kernion, J., Lovitt, L., Ndousse, K., Amodei, D., Brown, T., Clark, J., Kaplan, J., McCandlish, S., and Olah, C. A mathematical framework for transformer circuits. Transformer Circuits Thread, 2021.

Geiger, A., Lu, H., Icard, T., and Potts, C. Causal abstractions of neural networks. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2021.

Heimersheim, S. and Nanda, N. How to use and interpret activation patching. arXiv:2404.15255, 2024.

Helbling, A. et al. ConceptAttention: Diffusion transformers learn highly interpretable features. In *International Conference on Machine Learning (ICML)*, 2025. arXiv:2502.04320.

Hertz, A., Mokady, R., Tenenbaum, J., Aberman, K., Pritch, Y., and Cohen-Or, D. Prompt-to-prompt image editing with cross-attention control. In *International Conference on Learning Representations (ICLR)*, 2023.

Ho, J. and Salimans, T. Classifier-free diffusion guidance. In *NeurIPS Workshop on Deep Generative Models and Downstream Applications*, 2022.

Ho, J., Jain, A., and Abbeel, P. Denoising diffusion probabilistic models. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.

Meng, K., Bau, D., Andonian, A., and Belinkov, Y. Locating and editing factual associations in GPT. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.

Nanda, N., Chan, L., Lieberum, T., Smith, J., and Steinhart, J. Progress measures for grokking via mechanistic interpretability. In *International Conference on Learning Representations (ICLR)*, 2023.

Olsson, C., Elhage, N., Nanda, N., Joseph, N., DasSarma, N., Henighan, T., Mann, B., Askell, A., Bai, Y., Chen, A., et al. In-context learning and induction heads. Transformer Circuits Thread, 2022.

Orgad, H., Kavar, B., and Belinkov, Y. Editing implicit assumptions in text-to-image diffusion models. In *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2023.

Peebles, W. and Xie, S. Scalable diffusion models with transformers. In *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2023.

Power, A., Burda, Y., Edwards, H., Babuschkin, I., and Misra, V. Grokking: Generalization beyond overfitting on small algorithmic datasets. arXiv:2201.02177, 2022.

Raffel, C., Shazeer, N., Roberts, A., Lee, K., Narang, S., Matena, M., Zhou, Y., Li, W., and Liu, P. J. Exploring the limits of transfer learning with a unified text-to-text transformer. *Journal of Machine Learning Research (JMLR)*, 2020.

Rombach, R., Blattmann, A., Lorenz, D., Esser, P., and Ommer, B. High-resolution image synthesis with latent diffusion models. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022.

Tang, R., Liu, L., Pandey, A., Jiang, Z., Yang, G., Kumar, K., Stenetorp, P., Lin, J., and Ture, F. What the DAAM:

Interpreting stable diffusion using cross attention. In *Annual Meeting of the Association for Computational Linguistics (ACL)*, 2023.

Wang, B., Fan, J., and Pan, X. Circuit mechanisms for spatial relation generation in diffusion transformers. arXiv:2601.06338, Kempner Institute, Harvard University, 2026.

Wang, K., Variengien, A., Conmy, A., Shlegeris, B., and Steinhardt, J. Interpretability in the wild: a circuit for indirect object identification in GPT-2 small. In *International Conference on Learning Representations (ICLR)*, 2023.

Zarei, A., Basu, S., et al. Localizing knowledge in diffusion transformers. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2025. arXiv:2505.18832.

Zhang, F. and Nanda, N. Towards best practices of activation patching in language models: metrics and methods, 2024.

A. Experimental setup details

Architecture. Cross-attention has the conventional form

$$\text{Attn} = \text{softmax}\left(\frac{XW_Q(EW_K)^\top}{\sqrt{d_{\text{head}}}}\right)EW_V, \quad (2)$$

with $X \in \mathbb{R}^{256 \times 384}$ (spatial latent tokens on a 16×16 grid), $E \in \mathbb{R}^{L_{\text{tok}} \times 4096}$ (T5 embeddings, $L_{\text{tok}} \approx 15$), $d=384$, $d_{\text{head}}=64$, $H=6$ heads/layer, $L=6$ layers. Training: 4000 epochs with EMA, CFG conditioning-dropout rate 0.1.

Held-out sets. (i) *Unseen color $\times$ shape*: $n=12$ prompts using colour/shape combinations not seen during training. (ii) *Synonym-paraphrased relations*: $n=7$ prompts substituting synonyms (*on top of*, *underneath*, *above and to the left of*). (iii) Harder OOD: see Appendix H.

OpenCV evaluator. For each 128×128 image, we threshold to a binary mask, find external contours via `findContours` (RETR_EXTERNAL), classify each contour by mean RGB inside the contour (within margin 25 of canonical red/blue) and by polygon-segment count from `approxPolyDP` ($3 \rightarrow$ triangle, $4 \rightarrow$ square, $\geq 7 \rightarrow$ circle). For an image to score, both objects must be detected with the right colour and shape and their centroids' relative offset ($\Delta x, \Delta y$) must satisfy the relation. *Strict*: $|\Delta x|, |\Delta y| \geq 5$ pixels. *Loose*: ≥ 8 pixels with the right sign for the relation (e.g. *upper_right* requires $\Delta x \geq 8$ and $\Delta y \leq -8$). *Strict* additionally requires no spatial overlap.

Inference settings. DPM-Solver multi-step scheduler at 14 inference steps, CFG scale 4.5 (matching training), 10 images per prompt per condition. Pilot single-step ablations produced < 2 pp change, so we report all-step ablation throughout.

B. Candidate-head discovery details

Spatial-ramp basis. The basis consists of $D=8$ relation directions $\mathcal{D} = \{\text{above, below, left, right, upper_left, upper_right, lower_left, lower_right}\}$. For each r with unit vector $(d_y^{(r)}, d_x^{(r)}) \in \{-1, 0, +1\}^2$, the ramp $\mathbf{R}_r \in \mathbb{R}^{64}$ over the 8×8 patch grid is $[\mathbf{R}_r]_{(i,j)} = d_y^{(r)} \tilde{y}_i + d_x^{(r)} \tilde{x}_j$ with $\tilde{y}_i, \tilde{x}_j \in \text{linspace}(-1, 1, 8)$, mean-centred. The latent has 16×16 patches but the cross-attention output factorises within 8×8 super-tiles, so we operate at 8×8 .

Effect-vector decomposition. For each prompt embedding $\mathbf{e}_i \in \mathbb{R}^{4096}$ we run an additive-effects regression over factors $\mathcal{F} = \{\text{rel, color}_1, \text{color}_2, \text{shape}_1, \text{shape}_2, \text{interactions}\}$:

$$\mathbf{e}_i = \boldsymbol{\mu} + \sum_{f \in \mathcal{F}} \alpha_{f,i} \mathbf{v}_f + \boldsymbol{\epsilon}_i. \quad (3)$$

Pushing $\mathbf{v}_{\text{rel}}$ through the caption projection gives $\mathbf{p}_{\text{rel}} = W_{\text{cap}} \mathbf{v}_{\text{rel}} \in \mathbb{R}^{384}$. The variance partition reports $R_f^2 = 1 - \text{SS}_{\text{resid, w/o } f} / \text{SS}_{\text{total}}$.

Permutation null. 200 shuffled-relation refits give a null distribution that peaks at $|p|=8.3$ with mean 4.1 ± 1.6 . $L_0 H_0$'s observed $|p|=47.7$ is a $\sim 27\sigma$ departure ($p < 10^{-4}$). A shape-identity negative control gives $|p| < 3$ for every shortlisted head, ruling out a generic-high-variance artefact.

C. Causal ablation details

Zero-head ablation. With the per-head attention output tensor of shape $(B, H, P, d_{\text{head}})$ where B is the CFG-doubled batch ($B=2$), $H=6$ heads, $P=256$ image patches, $d_{\text{head}}=64$, we set `tensor[:,  $h_{\text{target}}$ , :, :] = 0` at every denoising step and on both CFG branches. The per-head output is zeroed *before* concatenation and projection through W_O .

Norm-matched random control. Five non-shortlist heads of comparable OV Frobenius norm. Mean ablation effect -0.003 ± 0.011 pp (95% CI excludes $L_0 H_0$ at $> 40\sigma$).

D. Downstream-reader search details

Method definitions.

- M1 (**offline OV–QK alignment**): rank by $|\cos|$ between candidate’s OV write and the spatial-ramp basis.
- M2 (**path-patching recovery loss**): from a clean / relation-flipped pair, patch L_0H_0 from clean (restoring attention) and additionally re-zero h_d ; the recovery loss measures h_d ’s on-path causal contribution.
- M3 (**in-vivo activation correlation**): capture every head’s per-step head-output for 30 prompts, compute per-prompt Pearson correlation between L_0H_0 and h_d , average across prompts.
- M4 (**per-head spatial-ramp projection**): per-position ℓ_2 norm of h_d ’s head-output (averaged over time and head.dim) correlated with the canonical 8×8 ramp R_r for the prompt’s relation; report mean $|\cos|$ across prompts.

For each method we rank the 35 non- L_0H_0 heads. Path-patching does not sweep L_0 , so L_0 entries get the column median. Consensus is the unweighted mean rank.

Table 8. Cross-method consensus ranking. Each cell is the head’s rank in that method (1 = best); consensus is the mean rank.

Cand.	M1	M2	M3	M4	Cons.
L_1H_2	2	11	7	2	5.50
L_0H_5	5	11	3	6	6.25
L_1H_3	6	1	10	17	8.50
L_2H_1	8	3	12	14	9.25
L_2H_2	14	3	20	1	9.50
L_2H_3	19	32	23	12	21.50
L_3H_1	15	3	16	30	16.00
L_4H_2	26	3	14	21	16.00

Activation-patching protocol. Patches are applied at the per-head attention output (post-OV, pre-projection) of h_d , on the conditional half of the CFG batch only, at all 14 denoising steps. Eight relation-flip pairs: above $\leftrightarrow$ below, left $\leftrightarrow$ right, and the four diagonals.

E. Reconciling the patching null with pair-ablation

L_1H_2 has a positive pair-ablation interaction ($I = +0.086$, CI excludes zero) but a 0% activation-patching recovery. Three explanations are consistent with this asymmetry:

- **Redundancy.** L_1H_2 writes a partially-redundant copy of L_0H_0 ’s output; in the corrupt forward pass L_0H_0 already injects the flipped pattern into the residual, so patching L_1H_2 from clean adds the same direction without overriding it. Joint ablation eliminates both copies at once.
- **Interaction-only contribution.** If L_1H_2 contributes only when the source-pair output exceeds a threshold, single-head patching cannot trigger it. The pair-ablation interaction is positive because L_1H_2 ’s contribution becomes detectable only when L_0H_0 is removed.
- **Patching-site mismatch.** We patch at the per-head attention output (post-OV, pre-projection); a more sensitive site might be the residual stream after the layer’s output projection.

Discriminating among these requires denser patching experiments (residual-stream patching, scaled patching, partial-replacement patching), left for future work.

F. Metric-resolved pair-ablation table

G. Self-attention prior under full cross-attention ablation

Ablating all 36 cross-attention heads collapses spatial accuracy to 2% but preserves $\sim 90\%$ colour and $\sim 30\%$ shape (Table 4, last row). This surviving generation comes from self-attention, trained during CFG-unconditional steps ($p(\text{img})$, no text) to produce coherent shaped objects without any text conditioning. Relational information enters only via cross-attention, so

Table 9. Metric-resolved pair-ablation. Δ = baseline – ablated; positive = damage. 30-prompt held-out set.

Condition	Δ_{rel}	Δ_{col}	Δ_{sh}	Δ_{bind}
L_0H_0 alone	+0.483	+0.147	+0.263	+0.273
$\{L_0H_0, L_1H_2\}$	+0.640	+0.180	+0.270	+0.310
$\{L_0H_0, L_0H_5\}$	+0.480	+0.190	+0.267	+0.293
$\{L_0H_0, L_1H_3\}$	+0.500	+0.210	+0.283	+0.297
$\{L_0H_0, L_2H_1\}$	+0.500	+0.173	+0.283	+0.300
$\{L_0H_0, L_2H_2\}$	+0.487	+0.140	+0.260	+0.267

removing it destroys the spatial signal but leaves the self-attention object prior intact. The two source heads project onto overlapping ramp directions (cosine $\rho \approx 0.7$ between their OV writes) but jointly cover more of the basis than either alone — the parallel redundancy is consistent with CFG dropout favouring such backup configurations.

H. Harder-OOD prompt variants

We built three harder OOD variants and re-encoded 24 prompts via T5-XXL: *distractor* (filler clause appended: “red square is above blue circle, in a quiet scene”), *extra adjectives* (“the small red square is above the large blue circle”), and *syntax-fronted* (“Above the blue circle is the red square”). The vanilla effect (+50 pp damage, baseline 0.83) reproduces the in-distribution result. On all three harder variants the model’s baseline drops to 0.38–0.42 before any ablation. L_0H_0 ablation produces additional damage on every variant (+7 to +14 pp), with relative drops $|\Delta|/\text{baseline}$ of 17% (*distractor*), 37% (*extra adjectives*), and 22% (*syntax-fronted*), versus 61% on vanilla. The circuit is not invariant to prompt syntax; its absolute effect is largest on the training distribution. This bounds the generalisation claim and identifies prompt-syntax robustness as a separate axis.

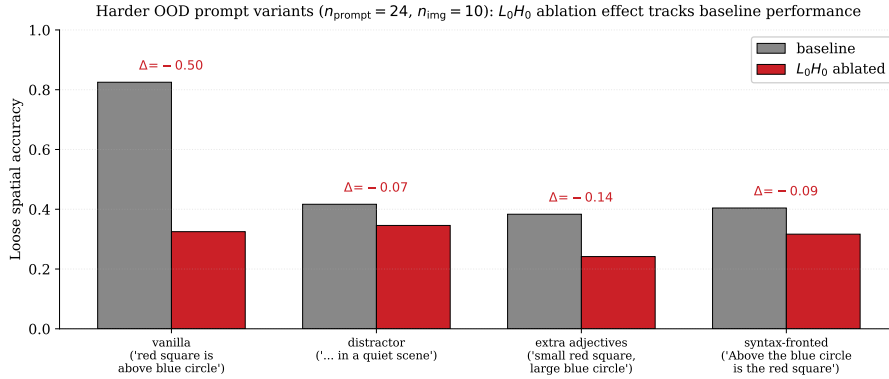


Figure 11. Harder OOD prompt variants. Vanilla template: L_0H_0 ablation causes +50 pp damage. The three harder variants degrade the model’s baseline accuracy substantially (from 0.83 to 0.38–0.42); L_0H_0 ’s marginal contribution becomes correspondingly smaller in absolute terms (+7 to +14 pp).

I. Triple-ablation bar chart

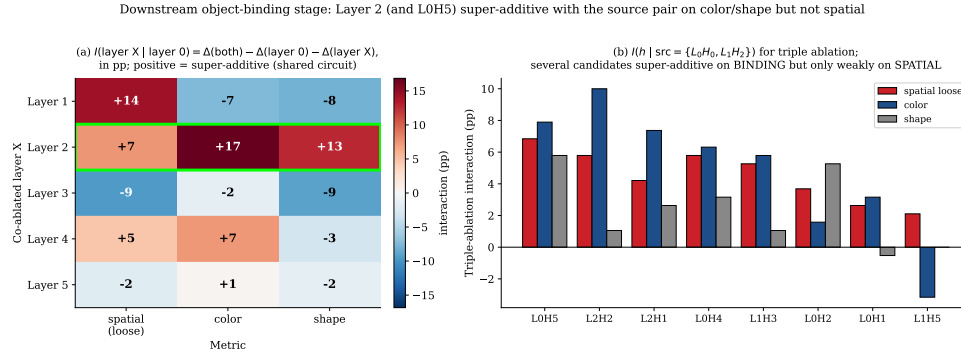


Figure 12. (a) Layer-pair interaction with layer 0; layer 2 (lime) shows positive interaction on colour and shape, but only a smaller interaction on spatial accuracy. (b) Triple-ablation interaction $I(h \mid \{L_0H_0, L_1H_2\})$. Spatial interactions (red bars) are systematically smaller than the corresponding colour interactions (blue bars).